Packet Tracer – Wireless Enterprise LAN Using WRT300N Wireless Routers

Objectives

Part 1: Build an Enterprise Wireless Topology

Part 2: Configure Wired Infrastructure

Part 3: Configure WRT300N Wireless Routers

Part 4: Configure WPA2 Wireless Security

Part 5: Configure DHCP Services

Part 6: Connect Wireless Clients

Part 7: Verify Wireless Connectivity

Part 8: Test Wireless Roaming

Background / Scenario

You are the network administrator for a growing company that is expanding its wireless infrastructure.

Employees require continuous wireless access while moving throughout the office.

Because Cisco Packet Tracer 9.0.0 has limited enterprise wireless support, you will use two WRT300N Wireless Routers configured as Access Points.

Both wireless devices will advertise the same SSID and use the same WPA2 security settings, allowing clients to roam between coverage areas while remaining connected.

This lab demonstrates:

- Enterprise wireless deployment
- Multiple wireless access points
- WPA2 wireless security
- DHCP addressing
- Wireless client connectivity
- Basic roaming concepts

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 1 Router (2911)
- 1 Switch (2960)
- 2 Wireless Routers (WRT300N)
- 1 Server
- 2 Laptops

Rename:

- Router0 → R1
- Switch0 → SW1
- WirelessRouter0 → AP-1
- WirelessRouter1 → AP-2
- Server0 → DHCP-SERVER
- Laptop0 → LAPTOP-A
- Laptop1 → LAPTOP-B

Step 2: Connect Devices

Use Copper Straight-Through cables.

Device	Interface	Device	Interface
R1	G0/0	SW1	Fa0/24
DHCP-SERVER	Fa0	SW1	Fa0/1
AP-1	Ethernet 1	SW1	Fa0/2

Device	Interface	Device	Interface
AP-2	Ethernet 1	SW1	Fa0/3

Laptops connect wirelessly.

Part 2: Configure IP Addressing

Configure Router

On R1:

enable

configure terminal

interface g0/0

ip address 192.168.50.1 255.255.255.0

no shutdown

end

Configure DHCP Server

Click DHCP-SERVER

Desktop → IP Configuration

Enter:

Setting	Value
IP Address	192.168.50.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.50.1

Part 3: Configure DHCP Services

Open:

Services → DHCP

Turn DHCP ON

Create Pool:

Setting	Value
Pool Name	WLAN
Default Gateway	192.168.50.1
DNS Server	8.8.8.8
Start IP Address	192.168.50.100
Subnet Mask	255.255.255.0
Maximum Users	100

Click Add.

Part 4: Configure AP-1 (WRT300N)

Click AP-1

GUI → Setup

Configure:

Setting	Value
Local IP Address	192.168.50.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.50.1

Disable DHCP Server:

DHCP Server = Disable

Save Settings

Configure Wireless

GUI → Wireless

Setting	Value
Network Name (SSID)	COMPANY-WIFI

Save Settings

Configure Security

Wireless → Wireless Security

Setting	Value
Security Mode	WPA2 Personal
Passphrase	SecureWiFi123

Save Settings

Part 5: Configure AP-2 (WRT300N)

GUI → Setup

Configure:

Setting	Value
Local IP Address	192.168.50.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.50.1

Disable DHCP Server

Save Settings

Configure Wireless

Wireless → Basic Wireless Settings

Setting	Value
SSID	COMPANY-WIFI

Save Settings

Configure Security

Wireless → Wireless Security

Setting	Value
Security Mode	WPA2 Personal
Passphrase	SecureWiFi123

Save Settings

Why Use the Same SSID?

Wireless roaming requires:

- Same SSID
- Same security type
- Same password

Clients automatically associate with the strongest available signal.

Part 6: Configure Wireless Clients

Note: Do not forget to insert the WPC300N module

Configure LAPTOP-A

Desktop → PC Wireless

Select:

COMPANY-WIFI

Connect

Password:

SecureWiFi123

Obtain DHCP Address

Separate the DHCP-SERVER into a VLAN (Check if applicable

Desktop → IP Configuration

Select DHCP

Expected:

Item	Value
IP Address	192.168.50.100+
Subnet Mask	255.255.255.0
Gateway	192.168.50.1

Configure LAPTOP-B

Repeat the same process.

Connect to:

COMPANY-WIFI

Password:

SecureWiFi123

Obtain address using DHCP.

Part 7: Verify Connectivity

Test 1: Ping Router

From LAPTOP-A:

ping 192.168.50.1

Expected:

4 successful replies.

Test 2: Ping DHCP Server

ping 192.168.50.10

Expected:

4 successful replies.

Test 3: Ping Between Wireless Clients

From LAPTOP-A:

ping (depends on the provided ip address of the dhcp)

Expected:

Reply received.

From LAPTOP-B:

ping (depends on the provided ip address of the dhcp)

Expected:

Reply received.

Part 8: Test Wireless Roaming

Initial Test

Place LAPTOP-A near AP-1.

Verify:

ping 192.168.50.1

Replies should continue.

Simulate Roaming

Switch Packet Tracer to Physical Mode.

Move LAPTOP-A:

- Farther from AP-1
- Closer to AP-2

Packet Tracer may reassociate the laptop with AP-2.

Verify Connectivity

Run:

ping 192.168.50.1

Expected:

Connectivity remains available.

Note:

Packet Tracer roaming behavior is simplified and may not perfectly emulate real enterprise wireless roaming.

Troubleshooting Tips

If clients cannot connect:

✓ Verify SSID

COMPANY-WIFI

✓ Verify WPA2 password

SecureWiFi123

✓ Verify DHCP Server is ON

✓ Verify WRT300N DHCP is OFF

✓ Verify Router G0/0 is up

show ip interface brief

✓ Verify Server IP

192.168.50.10

✓ Verify WRT300N Management IPs

AP-1 = 192.168.50.2

AP-2 = 192.168.50.3

✓ Verify laptops use DHCP

What You Learned

- How enterprise wireless networks operate
- How multiple access points extend wireless coverage
- How WPA2 secures wireless traffic
- How DHCP automatically assigns addresses
- How clients connect to SSIDs
- How roaming works conceptually between access points
- How WRT300N devices can simulate enterprise wireless infrastructure in Packet Tracer 9.0.0

Challenge Extensions (Optional)

Challenge 1: Guest Wireless Network

Create:

GUEST-WIFI

Use:

- Different SSID
- Different WPA2 password

Connect additional laptops.

Challenge 2: Additional Coverage Areas

Add:

- AP-3
- AP-4

Use the same SSID:

COMPANY-WIFI

Verify connectivity throughout the topology.

Challenge 3: Wireless VLAN Research

Research:

- Employee VLANs
- Guest VLANs
- Multiple SSIDs mapped to VLANs

Explain how enterprise wireless networks separate users.

Challenge 4: Signal Strength Observation

Move laptops around the Physical Workspace.

Observe:

- Signal strength
- Reassociation behavior
- Connectivity changes

Challenge 5: WPA3 Research

Research:

- WPA3-Personal
- WPA3-Enterprise

- SAE Authentication

Compare them with WPA2-Personal and identify security improvements.